

Math 241, F14
Exam 1

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	20	
2	20	
3	20	
4	20	
5	20	
Total	100	

Good Luck !

Problem 1. (20 points) Given two vectors $\mathbf{a} = \langle \sqrt{3}, 3, 2\sqrt{6} \rangle$ and $\mathbf{b} = \langle -\sqrt{3}, 0, \sqrt{6} \rangle$.
 (a) Find $|\mathbf{a}|$, $|\mathbf{b}|$, and $\mathbf{a} \cdot \mathbf{b}$.

$$|\mathbf{a}| = \sqrt{(\sqrt{3})^2 + 3^2 + (2\sqrt{6})^2} = \sqrt{3 + 9 + 24} = \sqrt{36} = 6$$

$$|\mathbf{b}| = \sqrt{(-\sqrt{3})^2 + 0^2 + (\sqrt{6})^2} = \sqrt{3 + 0 + 6} = \sqrt{9} = 3$$

$$\mathbf{a} \cdot \mathbf{b} = \sqrt{3} \cdot (-\sqrt{3}) + 3 \cdot 0 + 2\sqrt{6} \cdot \sqrt{6} = -3 + 0 + 12 = 9.$$

(b) Find a unit vector $\mathbf{u}$ which has the same direction as $\mathbf{b}$.

$$\mathbf{u} = \frac{\mathbf{b}}{|\mathbf{b}|} = \frac{1}{3} \langle -\sqrt{3}, 0, \sqrt{6} \rangle = \langle -\frac{\sqrt{3}}{3}, 0, \frac{\sqrt{6}}{3} \rangle$$

(c) Find the angle θ between $\mathbf{a}$ and $\mathbf{b}$.

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} = \frac{9}{3 \cdot 6} = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{3}$$

(d) Find the vector projection of $\mathbf{b}$ onto $\mathbf{a}$, $\text{proj}_{\mathbf{a}} \mathbf{b}$.

$$\begin{aligned} \text{proj}_{\mathbf{a}} \mathbf{b} &= \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|^2} \mathbf{a} = \frac{9}{6^2} \langle \sqrt{3}, 3, 2\sqrt{6} \rangle \\ &= \langle \frac{\sqrt{3}}{4}, \frac{3}{4}, \frac{\sqrt{6}}{2} \rangle \end{aligned}$$

Problem 2. (20 points) Given three points $P(-1, 3, 1)$, $Q(0, 5, -2)$ and $R(4, 3, -1)$.

(a) Find parametric equations and symmetric equations for the line through the points P and Q .

$$\begin{aligned} Y(t) &= \langle -1, 3, 1 \rangle + t(\langle 0, 5, -2 \rangle - \langle -1, 3, 1 \rangle) \\ &= \langle -1, 3, 1 \rangle + t\langle 1, 2, -3 \rangle = \langle -1+t, 3+2t, 1-3t \rangle \end{aligned}$$

Parametric equation: $x = -1+t, y = 3+2t, z = 1-3t$

symmetric equation: $\frac{x+1}{1} = \frac{y-3}{2} = \frac{z-1}{-3}$

(b) Find an equation for the plane that contains the triangle $\triangle PQR$.

$$\vec{PQ} = \langle 1, 2, -3 \rangle, \quad \vec{PR} = \langle 5, 0, -2 \rangle$$

$$\vec{PQ} \times \vec{PR} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -3 \\ 5 & 0 & -2 \end{vmatrix} = (-4-0)\hat{i} - (-2+15)\hat{j} + (0-10)\hat{k}$$

$$= -4\hat{i} - 13\hat{j} - 10\hat{k}$$

Thus the equation of the plane is given by.

$$-4(x+1) - 13(y-3) + (-10)(z-1) = 0$$

$$-4x - 13y - 10z = -45$$

(c) Find the area of the triangle $\triangle PQR$.

$$\text{Area}(\triangle PQR) = \frac{1}{2} |\vec{PQ} \times \vec{PR}|$$

$$= \frac{1}{2} |-4\hat{i} - 13\hat{j} - 10\hat{k}|$$

$$= \frac{1}{2} \sqrt{(-4)^2 + (-13)^2 + (-10)^2} = \frac{1}{2} \sqrt{16 + 169 + 100}$$

$$= \frac{1}{2} \sqrt{285}$$

Problem 3. (20 points)

(a) Let $\mathbf{r}(t) = 2t\mathbf{i} + t^2\mathbf{j} - \frac{1}{3}t^3\mathbf{k}$, find the arc length of the curve segment defined by $\mathbf{r}(t)$ with $0 \leq t \leq 1$.

$$\begin{aligned}\mathbf{r}'(t) &= 2\mathbf{i} + 2t\mathbf{j} - t^2\mathbf{k} \\ |\mathbf{r}'(t)| &= \sqrt{2^2 + (2t)^2 + (-t^2)^2} = \sqrt{4 + 4t^2 + t^4} = \sqrt{(t^2 + 2)^2} = t^2 + 2 \\ \Rightarrow L &= \int_0^1 |\mathbf{r}'(t)| dt = \int_0^1 t^2 + 2 dt = \left[\frac{t^3}{3} + 2t \right]_{t=0}^{t=1} \\ &= \left(\frac{1}{3} + 2 \right) - (0) \\ &= \frac{7}{3}\end{aligned}$$

(b) Let $\mathbf{r}(t) = \langle t, e^t \cos t, e^t \sin t \rangle$. Find the unit tangent vector $\mathbf{T}$ and the curvature κ of the curve at the point with $t = 0$.

$$\begin{aligned}\mathbf{r}'(t) &= \langle 1, e^t \cos t - e^t \sin t, e^t \sin t + e^t \cos t \rangle = \langle 1, e^t (\cos t - \sin t), e^t (\cos t + \sin t) \rangle \\ \mathbf{r}''(t) &= \langle 0, e^t (\cos t - \sin t) + e^t (-\sin t - \cos t), e^t (\cos t + \sin t) + e^t (-\sin t + \cos t) \rangle \\ &= \langle 0, -2e^t \sin t, 2e^t \cos t \rangle \\ \Rightarrow \mathbf{r}'(0) &= \langle 1, e^0 (\cos 0 - \sin 0), e^0 (\cos 0 + \sin 0) \rangle = \langle 1, 1, 1 \rangle \\ \Rightarrow \mathbf{T}(0) &= \frac{\mathbf{r}'(0)}{|\mathbf{r}'(0)|} = \frac{\langle 1, 1, 1 \rangle}{\sqrt{1^2 + 1^2 + 1^2}} = \langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \rangle \\ \mathbf{r}''(0) &= \langle 0, -2e^0 \sin 0, 2e^0 \cos 0 \rangle = \langle 0, 0, 2 \rangle \\ \mathbf{r}'(0) \times \mathbf{r}''(0) &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 0 & 0 & 2 \end{vmatrix} = 2\mathbf{i} - 2\mathbf{j} = \langle 2, -2, 0 \rangle \\ \Rightarrow \kappa(0) &= \frac{|\mathbf{r}'(0) \times \mathbf{r}''(0)|}{|\mathbf{r}'(0)|^3} = \frac{|\langle 2, -2, 0 \rangle|}{|\langle 1, 1, 1 \rangle|^3} = \frac{\sqrt{2^2 + (-2)^2 + 0^2}}{(\sqrt{3})^3} \\ &= \frac{\sqrt{8}}{3\sqrt{3}} = \frac{2\sqrt{6}}{9}\end{aligned}$$

Problem 4. (20 points)

(a) Find the velocity vector $\mathbf{v}(t)$ and the position vectors $\mathbf{r}(t)$ of a particle that has the acceleration $\mathbf{a}(t) = 2t\mathbf{i} + e^t\mathbf{j} + e^{-t}\mathbf{k}$ and initial conditions $\mathbf{v}(0) = \mathbf{j} - \mathbf{k}$ and $\mathbf{r}(0) = \mathbf{i}$.

$$\begin{aligned}\mathbf{v}(t) &= \mathbf{v}(0) + \int_0^t \mathbf{a}(u) du \\ &= \langle 0, 1, -1 \rangle + \int_0^t \langle 2u, e^u, e^{-u} \rangle du \\ &= \langle 0, 1, -1 \rangle + \langle t^2, e^t - 1, -e^{-t} + 1 \rangle \\ &= \langle t^2, e^t, -e^{-t} \rangle \\ \mathbf{r}(t) &= \mathbf{r}(0) + \int_0^t \mathbf{v}(u) du \\ &= \langle 1, 0, 0 \rangle + \int_0^t \langle u^2, e^u, -e^{-u} \rangle du \\ &= \langle 1, 0, 0 \rangle + \langle \frac{t^3}{3}, e^t - 1, e^{-t} - 1 \rangle \\ &= \langle \frac{t^3}{3} + 1, e^t - 1, e^{-t} - 1 \rangle\end{aligned}$$

(b) Find all **first** and **second** partial derivatives of the function $f(x, y) = 3x^4y^3 + e^{2xy}$.

$$\begin{aligned}f_x &= 12x^3y^3 + 2ye^{2xy} & f_y &= 9x^4y^2 + 2xe^{2xy} \\ f_{xx} &= 36x^2y^3 + 4y^2e^{2xy} \\ f_{yy} &= 18x^4y + 4x^2e^{2xy} \\ f_{xy} &= 36x^3y^2 + (2e^{2xy} + 2y \cdot 2xe^{2xy}) \\ &= 36x^3y^2 + 2e^{2xy} + 4xye^{2xy} \\ f_{yx} &= 36x^3y^2 + 2e^{2xy} + 4xye^{2xy}\end{aligned}$$

Problem 5. (20 points) Find the limit if it exists or show the limit does not exist.

(a) $\lim_{(x,y) \rightarrow (0,1/2)} e^{-x} \sin(\pi y)$

$$= e^{-0} \sin(\pi \cdot \frac{1}{2})$$

$$= 1$$

(b) $\lim_{(x,y) \rightarrow (1,0)} \ln \left(\frac{1+y^2}{x^2+xy} \right)$

$$= \ln \left(\frac{1+0}{1^2+1 \cdot 0} \right) = \ln 1 = 0$$

(c) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{2x^3 + y^3}$

When $(x,y) \rightarrow (0,0)$ along the x-axis, we have

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{2x^3 + y^3} = \lim_{(x,y) \rightarrow (0,0)} \frac{0}{2x^3 + 0} = 0$$

When $(x,y) \rightarrow (0,0)$ along the line $x=y$, we have

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{2x^3 + y^3} = \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \cdot x}{2x^3 + x^3} = \lim_{(x,y) \rightarrow (0,0)} \frac{x^3}{3x^3} = \frac{1}{3}$$

$\Rightarrow \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{2x^3 + y^3}$ does not exist since $0 \neq \frac{1}{3}$.

(d) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{2x^2 + y^2}$

$$0 \leq \frac{x^2 y}{2x^2 + y^2} \leq \left(\frac{x^2}{2x^2 + y^2} \right) y \leq \frac{y}{2}$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{y}{2} = 0, \quad \lim_{(x,y) \rightarrow (0,0)} 0 = 0$$

$$\Rightarrow \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{2x^2 + y^2} = 0$$